

Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_replay_solver_evolution.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	43.01344	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	77.6406	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1390.52414	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	0.0	13.06	n/a	136.14616	1	0	0	0.973861	14.91
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	313.89126	2	0	0	0.95232	15.31
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.064135	0.064135	547.9113	1	0	0	0.930514	15.07

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 43.01344.
- Accepted epoch count: 0.

- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 77.6406.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1390.52414.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.

- Worst held-out instances:
- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 0.0.
- Train penalized gap: 8.958333.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 13.06.
- Held-out runtime ms: 136.14616.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 1.616667.
- Worst held-out instances:
- X-n275-k28: feasible=False, penalized gap=15.7, errors=["NameError: name 'str' is not defined", 'missing 274 customers']
- X-n247-k50: feasible=False, penalized gap=14.3, errors=["NameError: name 'str' is not defined", 'missing 246 customers']
- X-n223-k34: feasible=False, penalized gap=13.1, errors=["NameError: name 'str' is not defined", 'missing 222 customers']
- X-n190-k8: feasible=False, penalized gap=11.45, errors=["NameError: name 'str' is not defined", 'missing 189 customers']
- X-n176-k26: feasible=False, penalized gap=10.75, errors=["NameError: name 'str' is not defined", 'missing 175 customers']

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 313.89126.
- Accepted epoch count: 2.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.

- Robustness dispersion across held-out structure families: 0.0.
- Worst held-out instances:
- X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.06703.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.064135.
- Held-out runtime ms: 547.9113.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.001517.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.098197, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.075066, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.053828, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.049565, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.044019, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (Conservative)

Condition	Feasibility (Heldout)	Feasible Gap	Penalized Gap	Runtime (ms)	Robustness Dispersion	Complexity	Code Novelty	Notes
Baseline Nearest Neighbor Constructive	100%	0.2734	0.2734	43	0.00278	0	0	Fastest baseline, moderate gap
Baseline Clarke-Wright Savings	100%	0.0738	0.0738	77.6	0.00779	0	0	Best fixed baseline for gap vs. runtime
Baseline Regret Insertion Local Search	100%	0.4664	0.4664	1390.5	0.0869	0	0	Worst gap, longest runtime
Direct Synthesis + One Repair	0%	0.0	13.06	136	1.6167	14.9	0.974	No feasibility heldout, high penalty & dispersion

Condition	Feasibility (Heldout)	Feasible Gap	Penalized Gap	Runtime (ms)	Robustness Dispersion	Complexity	Code Novelty	Notes
Budget-Matched Independent Search (No Replay)	0%	0.0	3.0	314	0	15.3	0.952	No feasibility, moderate penalty
Replay-Aware Iterative Search (Solver Evolution)	100%	**0.0641*	**0.0641**	548	**0.00152*	15.1	0.931	Best gap & robustness among learned methods, feasible

Interpretation

- **Feasibility:**
 - All fixed baselines achieved 100% feasibility on heldout data.
 - Learned methods without replay (direct generation + repair and budget-matched no-replay) failed to produce feasible solutions on heldout.
 - Replay-aware solver evolution matched fixed baselines in feasibility (100%).
- **Objective Gap:**
 - Replay-aware iterative search achieved the lowest heldout feasible gap (0.0641), slightly better than Clarke-Wright baseline (0.0738).
 - Nearest neighbor and regret insertion were worse.
 - Learned methods without replay exhibited very high penalized gaps due to infeasibility.
- **Runtime & Robustness:**
 - Nearest neighbor fastest, replay-aware iterative search moderate (548 ms), Clarke-Wright faster (77.6 ms).
 - Replay-aware method exhibited lowest robustness dispersion (0.00152), indicating consistent performance across structure families.
- **Complexity & Code Novelty:**
 - Learned conditions had higher complexity (~15) and novelty (~0.93–0.97).
 - Fixed baselines had zero complexity/novelty.
- **Replay vs. No Replay:**
 - Replay-aware iterative search clearly outperformed budget-matched no-replay in feasibility, gap, robustness, and produced feasible solutions.
 - Budget-matched no-replay failed feasibility despite similar complexity and novelty.

Conclusions

- Among learned methods, ****replay-aware iterative search (solver evolution)** clearly outperforms fixed baselines and budget-matched no-replay controls****** by achieving 100% feasibility, best objective gap, and strong robustness at moderate runtime.
- Learned conditions without replay failed to produce feasible solutions on heldout data despite reasonable runtime and novelty.
- ****Replay is critical**** for feasibility and quality in learned CVRP solution generation within this suite.
- Fixed baselines remain strong baselines with guaranteed feasibility and competitive gaps but are outperformed on gap by replay-aware iterative methods.